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CPU CONTROL UNIT DESIGN

4.1 Introduction

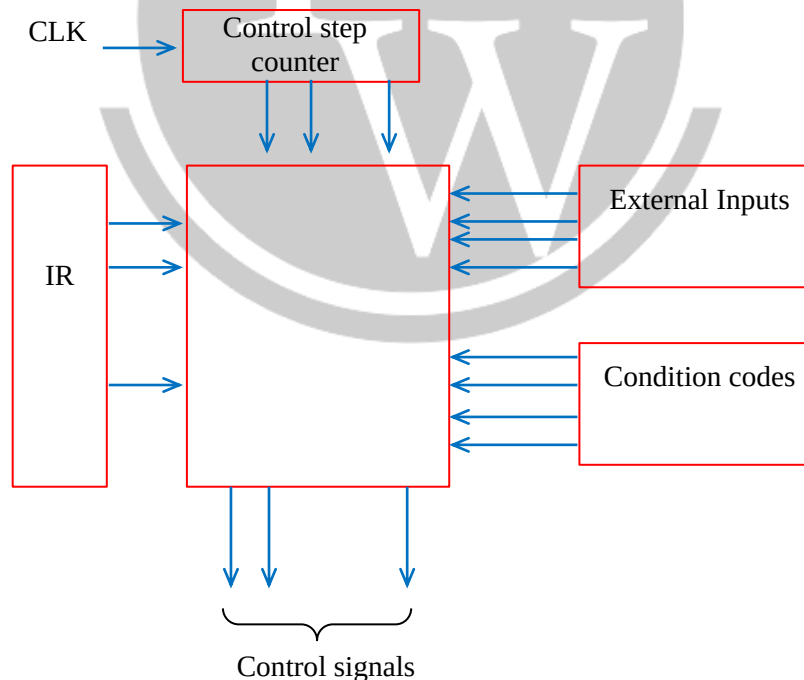
To execute instructions, the processor must have some means of generating the control signals needed in the proper sequence. The two approaches used for this purpose are

- (1) Hardwired control
- (2) Micro programmed control

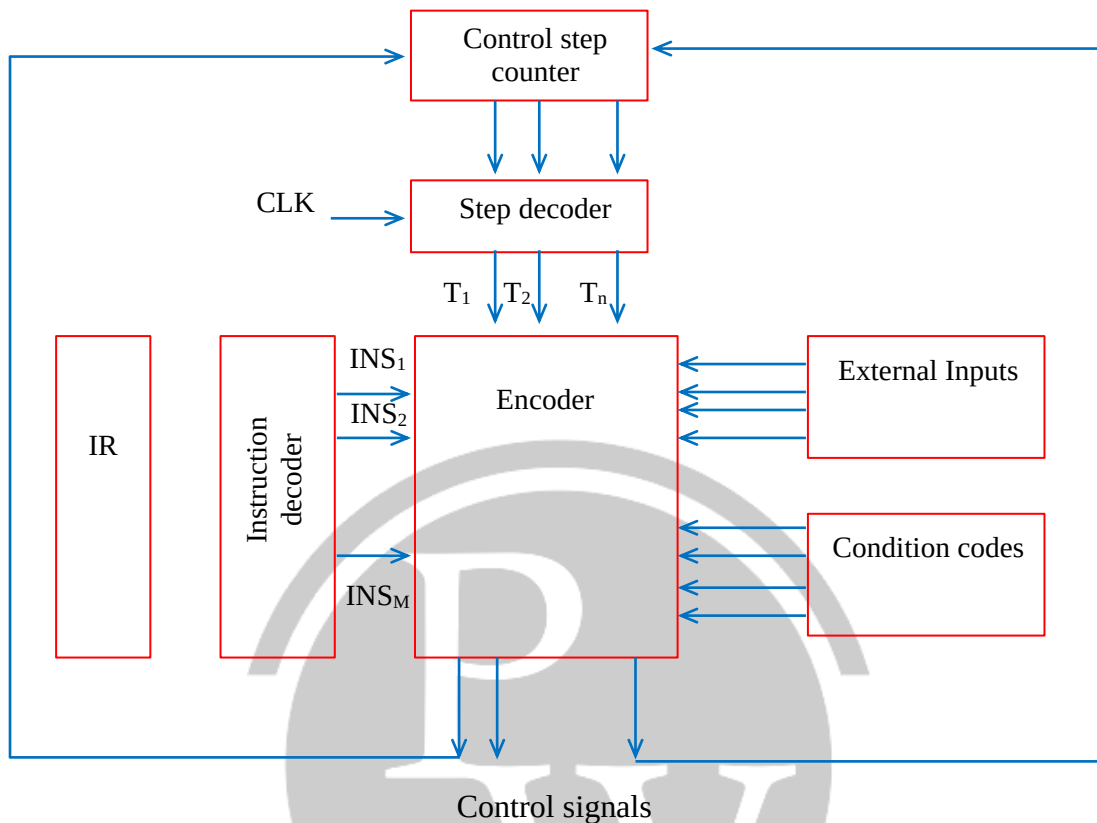
The purpose of control unit is to generate accurate timing & control signals.

4.1 Hardwired Control

- The control unit uses fixed logic circuits to interpret instructions and generate control signals from them. Every control signal is expressed as SOP (sum of products) expression and realized using digital hardware.



- The below figure shows the detailed hardwired control unit design.



- For executing an instruction completely each step is completed in one clock period. A counter is used to keep track of the control steps.
- The required control signals are determined with the following information.
 - (1) Contents of control step counter.
 - (2) Contents of instruction register
 - (3) Contents of condition code flags
 - (4) External input signals, such as MFC and interrupt requests.
 - The instruction decoder decodes the instruction loaded in IR
 - The step decoder provides a separate signal line for each step or time slot in a control sequence.
 - The encoder gets input from instruction decoder step decoder, external inputs and condition codes, thus uses to generate the individual control signals.
 - After execution of each instruction, the end signal is generated which resets control step counter and makes it ready for generation of control step for next instruction.

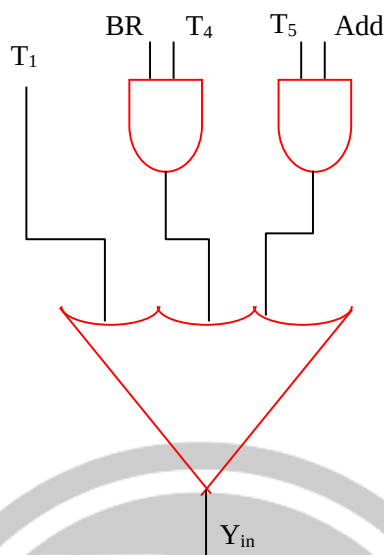
For example:

- (1) The encoder circuit implements the following logic function to generate Y_{in} .

$$Y_{in} = T_1 + T_5 \text{ Add} + T_4 \text{ BR} + \dots$$

i.e. Here Y_{in} signal is asserted during time interval T_1 for all instructions, during T_5 for add instruction, during T_4 for branch instruction.

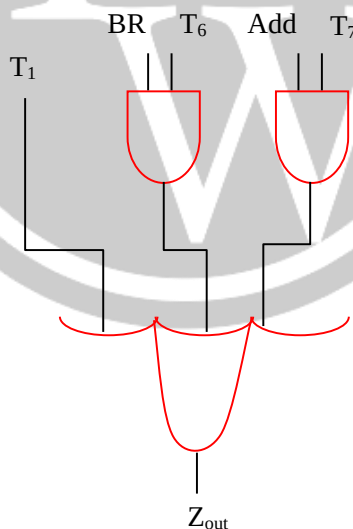
The generation of Y_{in} control signal is



(2) The logic function to generate Z_{out} signal can be given by

$$Z_{out} = T_2 + T_7 \text{ add} + T_6 \text{ BR} + \dots$$

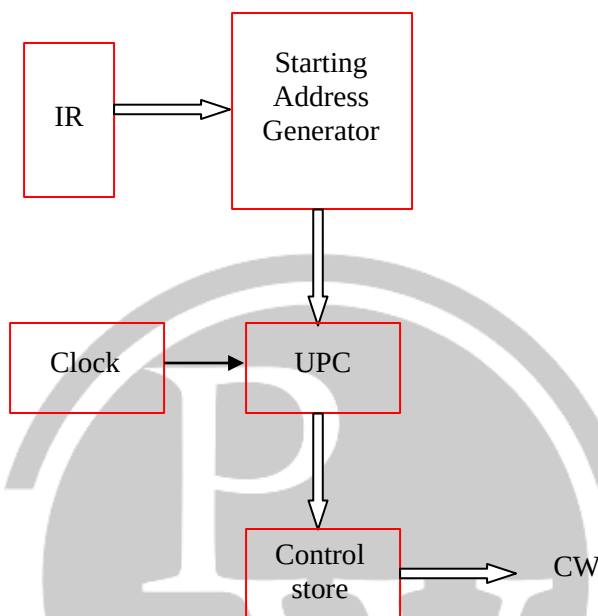
i.e., the Z_{out} signal is asserted during time interval T_2 for all instructions, during T_7 for add instruction, during T_6 for unconditional branch etc.



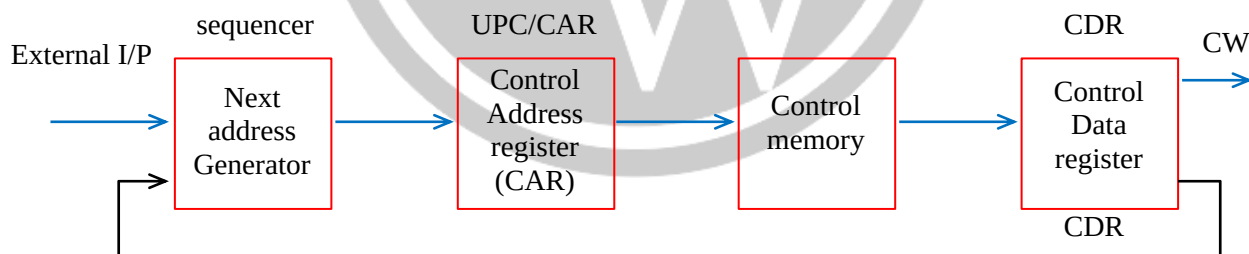
4.2 Microprogrammed Control

- Every instruction in a processor is implemented by a sequence of one or more sets of micro operations. Each micro operation is associated with a specific set of control lines which when activated, causes that micro operation to take place.
- Using micro programmed control, control signals are generated by a program. Using this the control signal selection and sequencing information is stored in ROM or RAM called control memory (CM).

- The control memory is part of control unit; it contains fixed programs (micro programs) that cannot be altered by occasional user.
- A control word (CW) is a word whose individual bits represent the various control signals.
- A sequence of CWs corresponding to the control sequence of a machine instruction constitutes the micro routine for that instruction, (micro program)
- The individual control words in the microprogram are referred as micro instruction.
- The basic organization of a microprogrammed control unit is

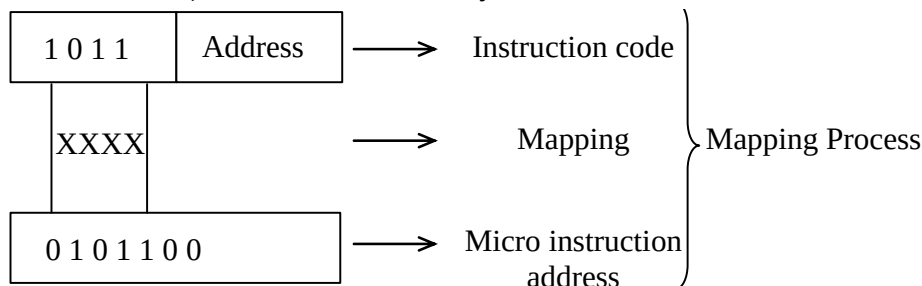


- To read the control words sequentially from the control memory a “micro program counter” (μ PC) is used.

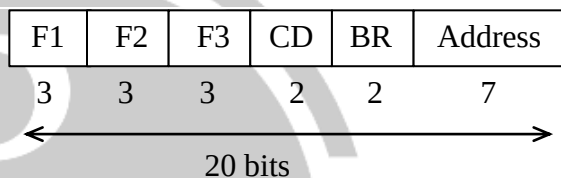


- The sequence of operations are:
 - (1) CAR holds the address of next micro instruction to be read.
 - (2) When address is available in UPC, the sequencer issues the READ command to CM.
 - (3) The word from addressed location is read into CDR/UIR.
 - (4) The UPC is incremented automatically by the clock, causing successive micro instructions to be read from CM.
 - (5) The contents of UIR generates control signals which are delivered to various parts of the processor in correct sequence.
- The μ PC or CAR can be updated with various options using the address sequencer circuit as:

- (1) When a new instruction is loaded in IR, the μ PC is loaded with the starting address of the micro routine for that instruction (called mapping process).
- (2) When a branch instruction is (micro) encountered and branch condition satisfied, the μ PC is loaded with the branch address.
- (3) When END instruction is encountered, the μ PC is loaded with address of first word.
- (4) In any other situation, the μ PC is incremented every time a new micro instruction is fetched from CM.



- The transformation from the instruction code bits to an address in control memory where the routine is located is called Mapping process.
- The basic microinstruction format is
- The function fields (F1, F2, F3,), condition field (CD) & Branch field (BR) may be optional.



Comparison between Hardwired & Microprogrammed

	Hardwired	Micro programmed
1. Speed	Fast	Slow
2. Control functions	Implemented in Hardwire	Implemented in software
3. Ability to handle complex instructions	Complex	Easier
4. Design process	Complicated	Orderly and systematic
5. Instruction set size	Under 100	Over 100
6. ROM size	NIL	2k to 10k (20–400 bit micro instruction)
7. Applications	RISC processors	CISC, Main frames

4.2.1. The Micro Programmed Control Unit Can be

(1) Horizontal Microprogramming:

- One bit per control signal.
- Maximum parallelism i.e., more than one signal can be simultaneously.
- No extra decoders are required for decoding.
- The length of control word is large, need to access more than once from control memory.



(2) Vertical Microprogramming:

- The control signals are encoded in form for k -bits 2^k signals.
 - Maximum degree of parallelism is 1.
 - The length of micro instruction is small.
 - Response is relatively slower.
 - Requires a decoder additionally.
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- To increase the degree of parallelism soft vertical microprogramming is used which divides the control signals into mutually exclusive groups. Each group is associated with associated number of control bits. (i.e., combination of both vertical and horizontal microprogramming).

